

TECHBIZ SECURITY

CYBER SECURITY INTERNSHIP PROGRAM

FOOTPRINTING & OSINT (PASSIVE RECONNAISSANCE)

Internship Task Report — Week 3, Task 7 (EH)

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Table of Contents

1. Introduction.....	3
2. Objectives.....	3
3. Footprinting.....	3
4. Open-Source Intelligence (OSINT).....	3
5. Passive Reconnaissance.....	4
6. Information Gathered During Passive Reconnaissance.....	4
7. Practical Lab - TryHackMe Passive Reconnaissance.....	5
8. DNSDumpster - Passive Subdomain Reconnaissance.....	5
9. Shodan - Passive Internet Exposure Reconnaissance.....	6
10. Google Dorking - OSINT Technique.....	7
11. TryHackMe Completion Evidence.....	8
12. Security Relevance of Footprinting and OSINT.....	8
13. Detection and Defensive Considerations.....	9
14. Key Learnings.....	9
15. Conclusion.....	9
16. Screenshot Evidence Index.....	9

1. Introduction

This report documents the completion of Week 3, Task 7 (EH) of the TechBiz Security Cyber Security Internship. The task focuses on understanding Footprinting, Open-Source Intelligence (OSINT), and Passive Reconnaissance, which together form an important early stage of the reconnaissance process in ethical hacking.

Reconnaissance is typically the first phase of a security assessment. Before any active testing takes place, security professionals gather information about a target's publicly visible presence to understand its attack surface. This includes domain information, DNS records, subdomains, exposed services, and other publicly indexed content.

Passive reconnaissance is particularly valuable at this stage because it allows information to be gathered from publicly available sources without directly interacting with, probing, or scanning the target's infrastructure. This task explores the concepts and tools used to perform passive reconnaissance and OSINT gathering in a controlled training environment.

2. Objectives

- Understand the concept of Footprinting.
- Understand Open-Source Intelligence (OSINT).
- Understand Passive Reconnaissance.
- Differentiate passive reconnaissance from active reconnaissance.
- Understand publicly available information that can be collected during reconnaissance.
- Learn how WHOIS can provide domain registration information.
- Understand how DNS information can be gathered using nslookup and dig.
- Understand passive subdomain discovery using DNSDumpster.
- Understand Certificate Transparency logs as a source of subdomain information.
- Understand how Shodan can provide information about publicly exposed services.
- Understand Google Dorking as an OSINT technique for discovering indexed information.
- Understand how reconnaissance findings can help both penetration testers and defenders.

3. Footprinting

Footprinting is the process of systematically collecting information about a target organisation's publicly visible presence, typically as the first step of a security assessment. The purpose of footprinting is to build a clear picture of what information about an organisation is available before any deeper or more direct testing is carried out.

Information that may be collected during footprinting includes domain and DNS information, IP addresses and general network information, subdomains, technologies and services in use, publicly available organisational information, and public documents or content indexed by search engines.

Footprinting is performed before further security testing because it allows a tester to understand the scope of what is publicly visible without immediately performing intrusive actions. The goal at this stage is not to exploit or directly test any system, but to build an understanding of the target's publicly visible attack surface using information that is already accessible.

4. Open-Source Intelligence (OSINT)

Open-Source Intelligence (OSINT) refers to the practice of collecting and analysing information that is available from publicly accessible sources. In a cybersecurity context, OSINT is a key part of the reconnaissance process, as it allows information to be gathered without requiring direct access to a target's systems.

Common OSINT sources include search engines, WHOIS registration records, DNS records, Certificate Transparency logs, dedicated DNS reconnaissance platforms such as DNSDumpster, internet-connected device search engines such as Shodan, and publicly available websites and documents.

OSINT is valuable to a penetration tester because it can reveal useful information about a target's infrastructure and exposure before active testing begins. At the same time, OSINT is equally valuable to defenders, as it allows an organisation to see what information about itself is already publicly exposed and identify anything that may need to be reviewed or removed.

5. Passive Reconnaissance

Passive reconnaissance is the process of collecting information about a target from publicly available sources without directly interacting with the target's infrastructure in a way that performs active probing or scanning. It relies on information that has already been published, indexed, or recorded elsewhere, such as DNS records, WHOIS data, Certificate Transparency logs, and internet-wide scan platforms like Shodan.

This differs from active reconnaissance, which involves direct interaction with the target, such as port scanning or service enumeration, and is generally more detectable. The table below summarises the key differences between the two approaches.

Aspect	Passive Reconnaissance	Active Reconnaissance
Interaction with target	Minimal or indirect	Direct
Typical sources	WHOIS, DNS information, CT logs, search engines, Shodan	Port scanning, service enumeration, direct probing
Detection possibility	Generally lower	Generally higher
Purpose	Gather publicly available intelligence	Directly identify live systems/services

6. Information Gathered During Passive Reconnaissance

Passive reconnaissance can reveal several categories of information that contribute to an understanding of an organisation's external attack surface. The table below summarises the main categories relevant to this task.

Information Type	Example	Security Relevance
Domain information	WHOIS registration details	Reveals domain ownership and registration data
DNS information	Records retrieved via nslookup/dig	Reveals how a domain resolves and its associated infrastructure
Subdomains	Subdomains identified via DNSDumpster or CT logs	May reveal additional publicly exposed services or infrastructure
IP/network information	Hosting and network ranges	Helps map infrastructure associated with a domain
Hosting/provider information	Cloud or hosting provider names	Indicates where infrastructure is hosted
Publicly exposed services	Services identified via Shodan	Shows what is visible to anyone on the internet
Technology/service banners	Server and software banners	May reveal technologies in use
Publicly indexed documents	Content located via Google Dorking	May reveal unintentionally exposed information
Public organisational information	General publicly available content	Contributes to overall attack surface awareness

7. Practical Lab – TryHackMe Passive Reconnaissance

The practical component of this task was performed through the TryHackMe “Passive Reconnaissance” learning room. As shown in the provided evidence, the room was completed with a 100% completion indicator. The practical exercises covered passive reconnaissance concepts and tools, including DNSDumpster and Shodan, and the evidence demonstrates successful completion of the relevant exercises within this room.

8. DNSDumpster – Passive Subdomain Reconnaissance

DNSDumpster is a passive reconnaissance resource used to gather DNS-related information about a domain, including associated subdomains, hosting and network details, and observed service banners. Because this information is drawn from existing DNS and certificate data, DNSDumpster can be used without sending direct traffic to the target's infrastructure.

Subdomain discovery is a useful part of reconnaissance because subdomains can reveal additional, sometimes less-monitored parts of an organisation's infrastructure. Identifying subdomains can highlight publicly exposed services or systems that may not be immediately obvious from an organisation's main domain alone.

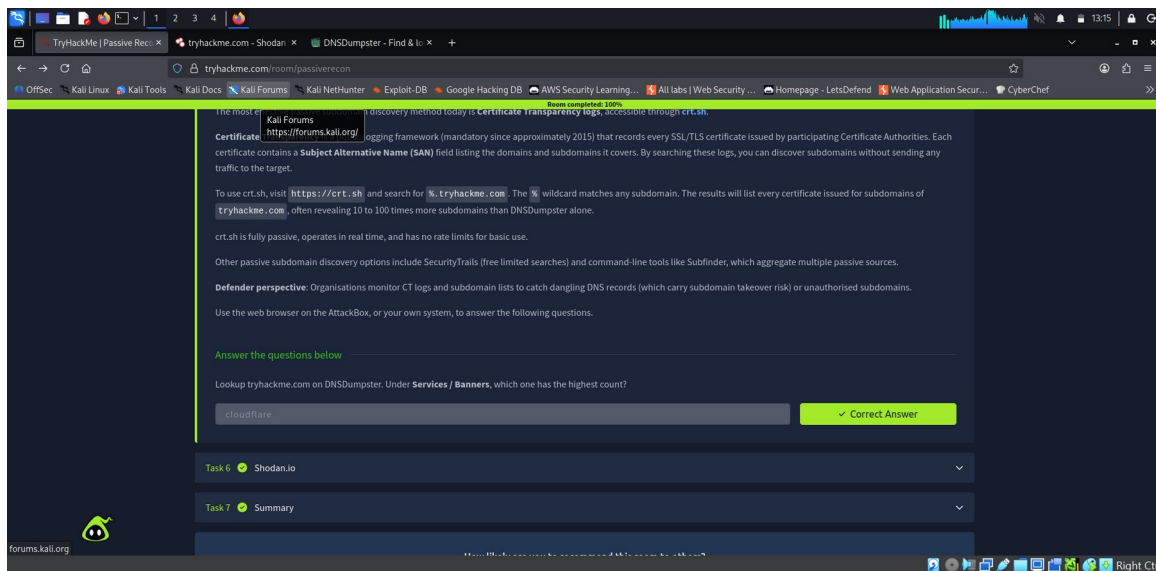


Figure 1 – DNSDumpster exercise showing the identified service/banner with the highest count.

As shown in Figure 1, the DNSDumpster exercise required identifying which service or banner had the highest count under the “Services / Banners” section for tryhackme.com, with “cloudflare” confirmed as the correct answer.

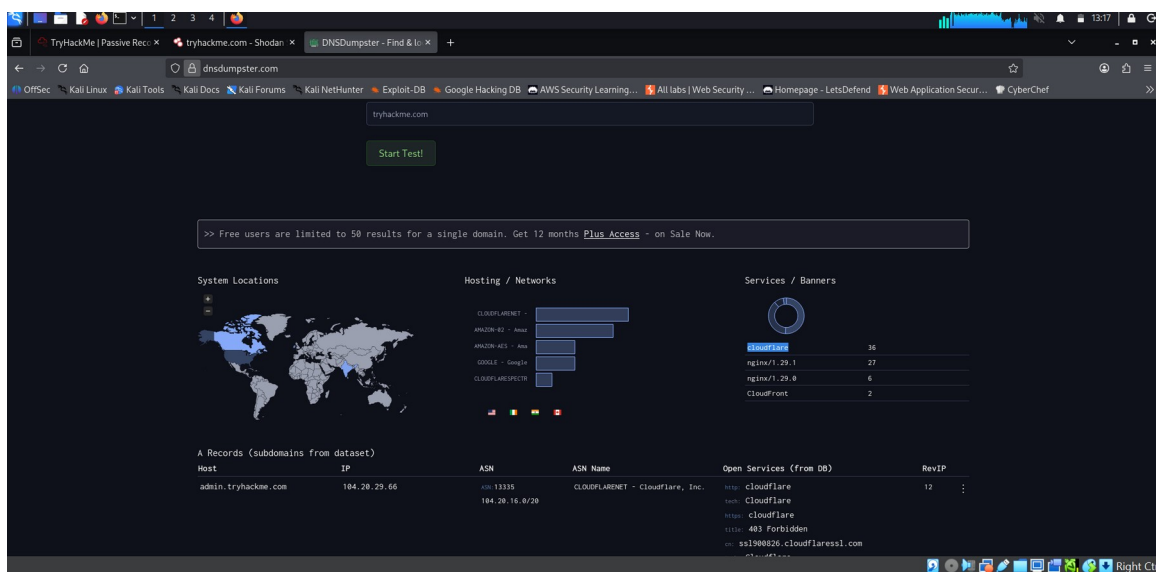


Figure 2 – DNSDumpster passive reconnaissance results showing domain-related infrastructure information.

Figure 2 shows the DNSDumpster results page for tryhackme.com, including System Locations, Hosting/Networks (such as Cloudflare, Amazon, and Google-associated ranges), Services/Banners (including cloudflare, nginx, and CloudFront entries), and an A Records table listing an identified subdomain (admin.tryhackme.com) along with its associated IP address, ASN information, and open services. This demonstrates how DNSDumpster can provide useful intelligence about a domain's infrastructure without performing direct active scanning against the target.

9. Shodan – Passive Internet Exposure Reconnaissance

Shodan is a search engine that indexes information about internet-connected systems. Rather than scanning a target directly, Shodan allows a user to search a database of previously collected information, making it a passive reconnaissance resource. Shodan can provide information such as publicly visible services, open ports, service banners, technologies in use, hosting providers, geographic information, and certificate-related metadata.

This type of information is valuable to defenders as well as testers, since it allows an organisation to see what an outside observer using Shodan could learn about its externally exposed systems without the organisation being directly involved in that lookup.

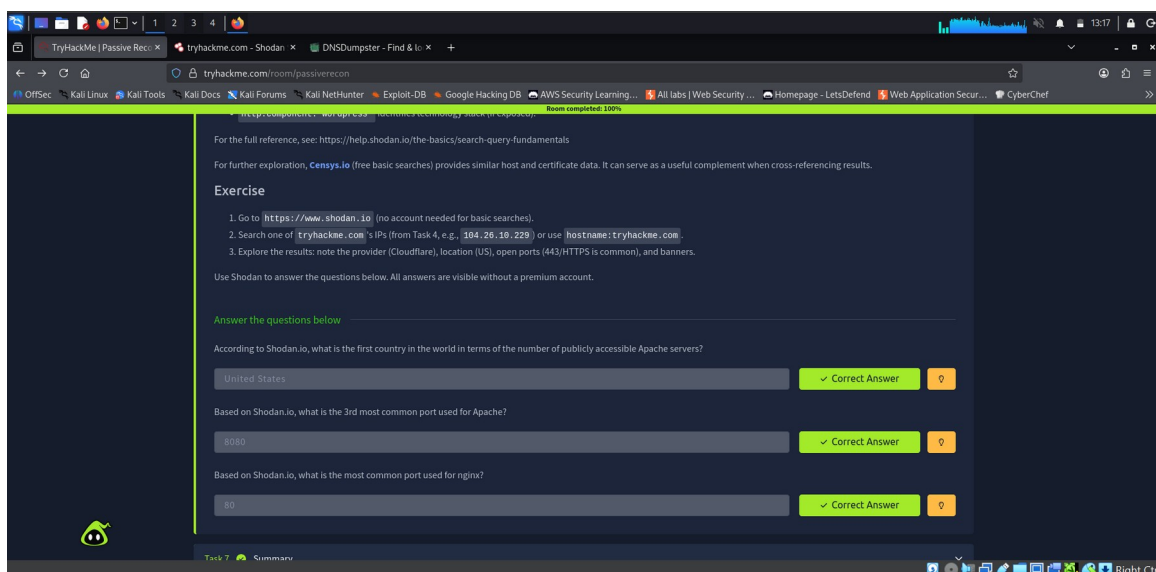


Figure 3 – TryHackMe Shodan exercises completed with correct answers.

Figure 3 shows the Shodan-related questions in the TryHackMe Passive Reconnaissance room completed successfully, covering the most common country for publicly accessible Apache servers, and the most common ports associated with Apache and nginx, all confirmed as correct answers.

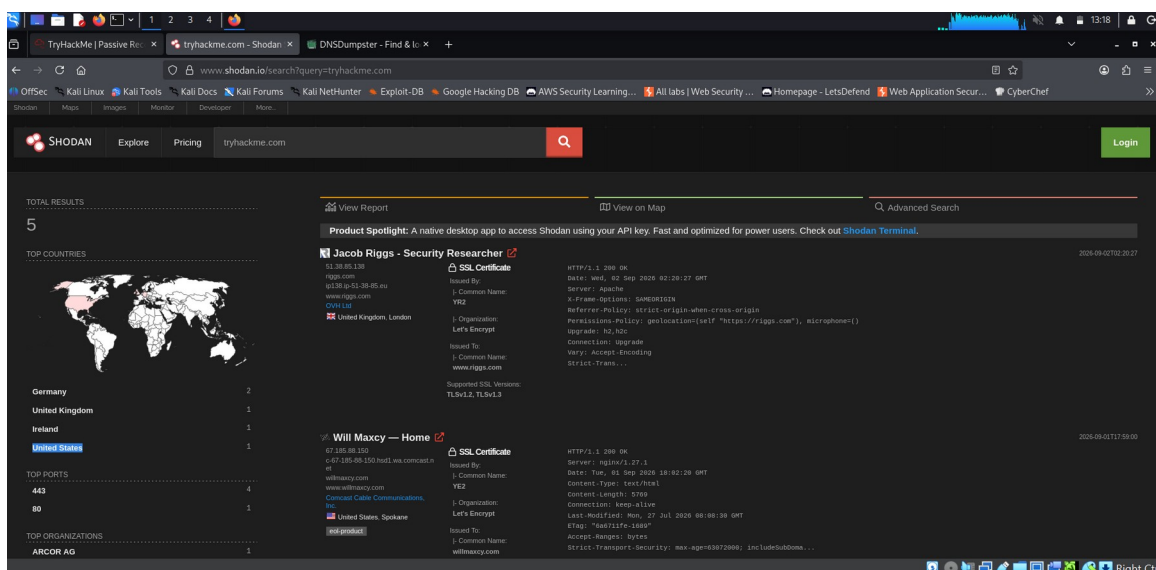


Figure 4 – Shodan search results demonstrating publicly visible service and infrastructure information.

Figure 4 shows Shodan search results for the query “tryhackme.com”, displaying five total results with associated countries (Germany, United Kingdom, Ireland, and United States), commonly observed ports (443 and 80), and organisation information, along with individual host entries showing details such as SSL certificate metadata, HTTP server headers, and issuing certificate authorities. This illustrates the type of publicly visible service and infrastructure information that can be collected through passive reconnaissance using Shodan. These results represent publicly indexed service information only and are not presented as evidence of any vulnerability.

10. Google Dorking – OSINT Technique

Google Dorking is an OSINT technique that uses advanced search engine operators to locate specific publicly indexed information that may not be easily found through a standard search. The practical exercise in this task covered three common operators: `site:`, which restricts search results to a specific domain; `filetype:`, which filters results by a specified file type; and `intitle:`, which searches for specific text within the title of indexed pages.

These operators can help security professionals identify publicly indexed content that may be relevant to a security assessment, such as specific file types or pages hosted on a particular domain. This technique is intended for locating information that is already publicly indexed and should not be used to attempt unauthorised access to sensitive or restricted content.

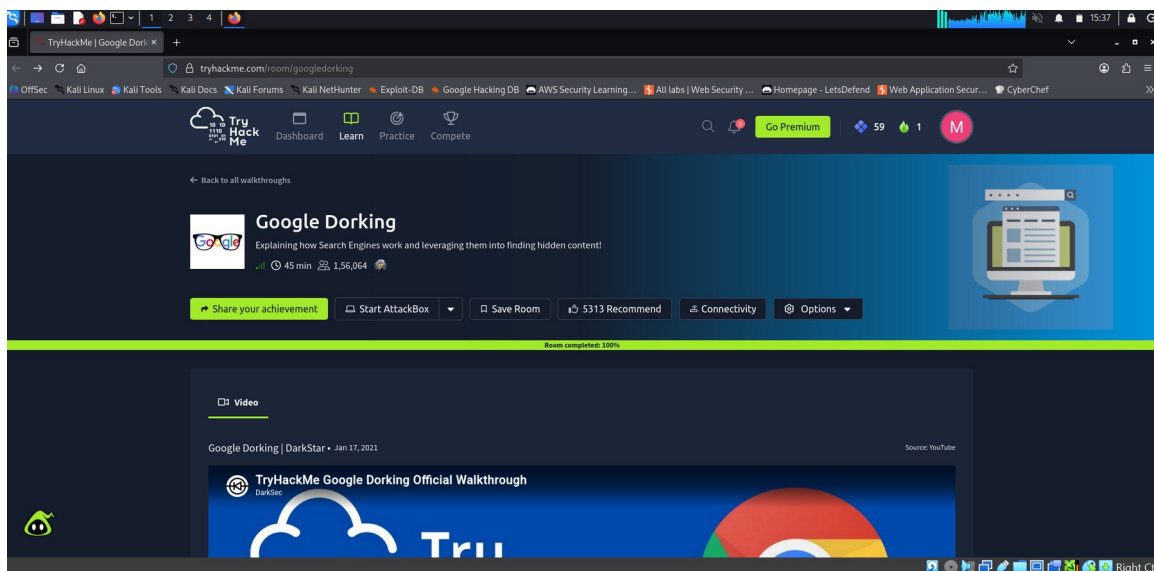


Figure 5 – TryHackMe Google Dorking room showing 100% completion.

Figure 5 provides evidence of successful completion of the Google Dorking learning room, with the room completion indicator showing 100%.

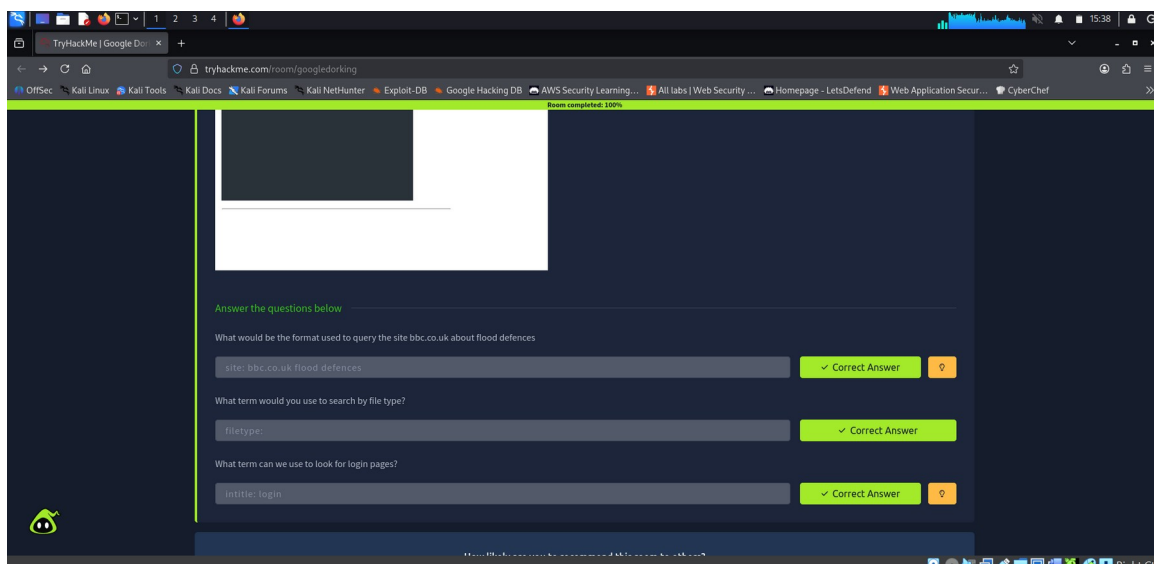


Figure 6 – Google Dorking exercises completed with correct answers.

Figure 6 shows the completed exercises for this room, covering the site: operator (used to query bbc.co.uk about flood defences), the filetype: operator (used to search by file type), and the intitle: operator (used to look for login pages), with each answer confirmed as correct. In a controlled, ethical hacking and OSINT context, these operators support the identification of publicly indexed information relevant to a security assessment.

11. TryHackMe Completion Evidence

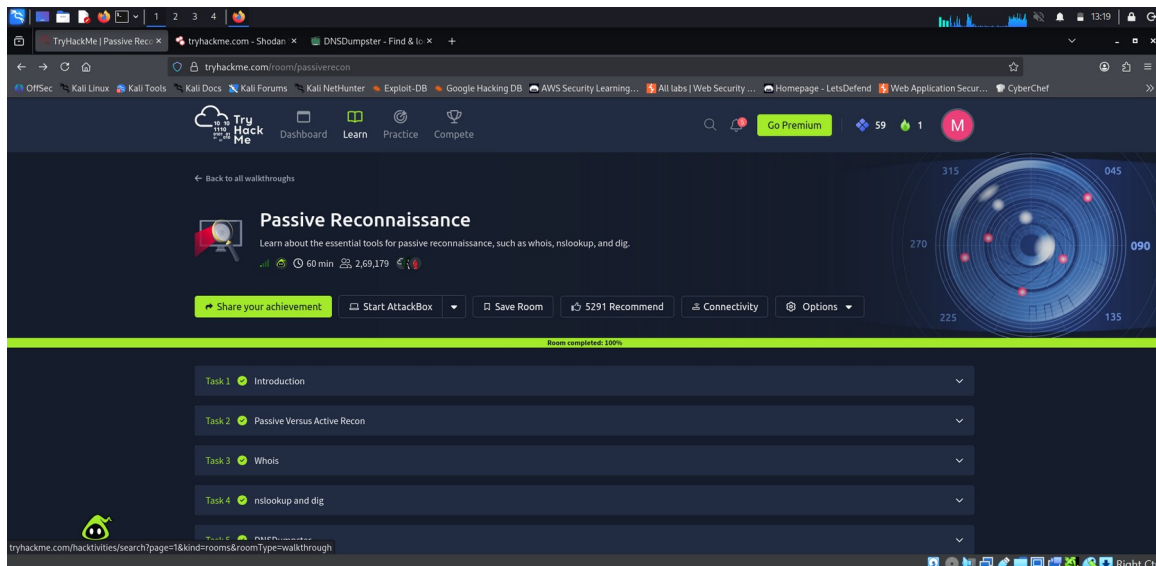


Figure 7 – TryHackMe Passive Reconnaissance room completed with 100% progress.

Figure 7 provides overall completion evidence for the Passive Reconnaissance room, showing the completed learning tasks, including Introduction, Passive Versus Active Recon, Whois, and nslookup and dig, alongside the 100% room completion indicator. No other TryHackMe room completion is claimed beyond what is supported by the provided screenshots.

12. Security Relevance of Footprinting and OSINT

Footprinting and OSINT are important early stages of ethical hacking because they support several key activities:

- Attack surface discovery.
- Identification of publicly exposed information.
- Subdomain discovery.
- Identification of publicly visible services.
- Understanding technology exposure.
- Information gathering before deeper security testing.
- Reducing unnecessary direct interaction with target infrastructure during initial reconnaissance.

From a defender's perspective, the same techniques and information sources can be used to strengthen an organisation's security posture:

- Monitor publicly exposed assets.
- Maintain accurate asset inventories.
- Remove unnecessary public information.
- Review DNS and subdomains.
- Monitor Certificate Transparency logs.
- Review internet-facing services.
- Search for accidentally exposed documents or information.

13. Detection and Defensive Considerations

Although passive reconnaissance generally produces less direct network interaction than active scanning, organisations should still monitor their external exposure, since the same publicly available information used during passive reconnaissance is available to any observer. Relevant defensive practices include:

- External attack-surface monitoring.
- DNS and subdomain inventory.
- Certificate Transparency monitoring.
- Monitoring publicly exposed services.
- Reviewing search engine indexing.
- Removing unnecessary public information.
- Monitoring cloud and hosting exposure.
- Regular security assessments.

14. Key Learnings

- Footprinting is an important reconnaissance stage.
- OSINT uses publicly available information.
- Passive reconnaissance reduces direct interaction with target infrastructure.
- WHOIS and DNS information can reveal useful domain intelligence.
- DNSDumpster can assist with passive subdomain discovery.
- Certificate Transparency can reveal certificate-associated domains.
- Shodan can reveal information about publicly accessible internet services.
- Google Dorking can identify publicly indexed information using search operators.
- Reconnaissance information helps security professionals understand an organisation's external attack surface.
- The same information can be used by defenders to identify and reduce unintended exposure.

15. Conclusion

This task provided practical understanding of Footprinting, OSINT, and Passive Reconnaissance as foundational stages of the ethical hacking process. Through the TryHackMe Passive Reconnaissance and Google Dorking learning rooms, practical exercises using DNSDumpster, Shodan, and Google Dorking operators were completed.

The exercises demonstrated how publicly available information can reveal domain, DNS, subdomain, service, hosting, and indexed-content information without relying on intrusive scanning. From an ethical hacking perspective, passive reconnaissance is an important initial phase for understanding an organisation's external attack surface before any authorised, more direct security testing is carried out.

16. Screenshot Evidence Index

No.	Screenshot Filename	Evidence
1	01_dnsdumpster_services_banners_answer.png	DNSDumpster exercise result
2	02_dnsdumpster_passive_subdomain_reconnaissance.png	DNSDumpster passive reconnaissance data
3	03_tryhackme_shodan_questions_completed.png	Shodan exercises completed
4	04_shodan_passive_reconnaissance_results.png	Shodan reconnaissance results
5	05_tryhackme_passive_reconnaissance_completed.png	Passive Reconnaissance room 100% completion
6	06_tryhackme_google_dorking_completed.png	Google Dorking room 100% completion
7	07_google_dorking_questions_completed.png	Google Dorking exercises completed